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New Silurian and Devonian palaeomagnetic results from the Hexi Corridor terrane, northwest China, and their tectonic implications

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SUMMARY

A total of 239 orientated drill-core samples from 23 sites were collected for palaeomagnetic study from Silurian and Devonian red beds, marlaceous sandstone, and limestone rocks in the eastern part of the Hexi Corridor, southwest Ningxia, North China. The characteristic high-temperature component resides in both haematite and magnetite. It clusters around a northwesterly and shallow to moderate downward direction and its antipode after tilt correction. The primary origin of this characteristic remanent magnetization (ChRM) is ascertained by positive fold and reversal tests at the 95 per cent confidence level. The corresponding palaeopoles, at 339.0°E, 60.1°N with $A_{95} = 11.2°$ (Silurian) and 336.0°E, 56.0°N with $A_{95} = 9.2°$ (Devonian), imply that the North China Block (NCB) had a low palaeolatitude of around 15°N in the Northern Hemisphere during the Silurian–Devonian period. Comparison with the Early–Middle Ordovician palaeopole of the NCB suggests that the NCB moved rapidly northwards by $30.8° \pm 10.9°$ to cross the palaeo-equator during the Early–Middle Ordovician to Silurian. In combination with the palaeobiogeographical data from Ningxia, our palaeomagnetic results suggest that the NCB was located close to Australia during the Late Devonian.

Key words: Devonian, North China Block, palaeomagnetism, Silurian.

1 INTRODUCTION

Palaeozoic palaeomagnetic results obtained from the three major blocks comprising China, namely North China, South China and Tarim, strongly favour the hypothesis that the Chinese blocks were located near the equator, and that they were part of East Gondwana or juxtaposed to East Gondwana during the Early Palaeozoic (Bai *et al.* 1987; Li 1990; Opdyke *et al.* 1987; Fang *et al.* 1989; Zhao *et al.* 1992, 1993, 1996; Van der Voo 1993; Yang *et al.* 1996). Palaeoclimatic and palaeobiogeographical data for the Chinese blocks also support this hypothesis (Audley-Charles *et al.* 1988; Long & Burrett 1989; Burrett *et al.* 1990; Nie 1991). Consistent with this, Huang *et al.* (1999a) recently speculated that the North China Block (NCB) was probably sited close to Antarctica, within East Gondwana, during the Early Cambrian, and may have episodically separated from Gondwana by the Middle Ordovician. However, the question of how the NCB connected with adjacent areas during the rest of the Cambrian–Devonian interval is a subject under debate. The extremely sparse nature of palaeomagnetic data prior to the Permian for the NCB, and in particular the lack of palaeomagnetic data in the Silurian–Devonian period, makes it very difficult to discuss its mid-Palaeozoic palaeogeographical connections in detail. This scarcity is due partially to the great stratigraphic hiatus in the NCB during the period from the Upper Ordovician to the Lower Carboniferous.

The severe lack of palaeomagnetic data during the Silurian–Devonian interval for the NCB currently precludes testing of the palaeogeographic models. It has recently been claimed that the Alashan/Hexi Corridor terrane, in which the Silurian–Devonian sediments are well developed, is the western extension of the NCB since the Late Cambrian (e.g. Zhao *et al.* 1993; Wu *et al.* 1993; Meng & Coe 1992). Consequently, a few palaeomagnetic studies spanning the Palaeozoic and Mesozoic period have recently been carried out in the Alashan/Hexi Corridor terrane, for the purpose of tectonic study to supplement the palaeomagnetic database from the NCB. For the Silurian and Devonian period, however, individual research groups present plausible but contradictory palaeomagnetic poles from this area (Li *et al.* 1985; Zhu & He 1987; He *et al.* 1988; Meng & Coe 1992; Zhao *et al.* 1993). These palaeomagnetic poles are distributed along a small circle centred on

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the eastern part of the Hexi Corridor. The Silurian–Devonian poles reported by Zhao *et al.* (1993) are from widely distributed localities, and pass fold and reversal tests. These palaeopoles are located far away from those of the Early and Late Palaeozoic for the NCB. Either true polar wander (TPW) or a simple local rotation could be involved. Because of the complicated strike-slip faults in the Hexi Corridor area, internal displacement and local deformation may have caused local rotation. Indeed, palaeomagnetic study from the Late Palaeozoic to the Mesozoic on the Alashan/Hexi Corridor reveals significant local rotation with respect to the NCB by the end of the Triassic, and the internal displacement and deformation might have occurred during the Mesozoic (Wu *et al.* 1993). Consequently, reliable Silurian and Devonian palaeopoles are required from the autochthonous terrane.

In this paper, we report a palaeomagnetic study of Silurian and Devonian rocks in the eastern part of the Hexi Corridor, southwest Ningxia, North China. By combining palaeontological data with new Silurian–Devonian palaeomagnetic results from the Alashan/Hexi Corridor terrane, the palaeogeographical position of the NCB will be examined.

2 GEOLOGICAL BACKGROUND AND PALAEOMAGNETIC SAMPLING

The Alashan/Hexi Corridor terrane, which is situated in the west of the Ordos Basin through the Helanshan fold belt, is at the juncture of two major Chinese blocks: North China and Tarim (Fig. 1a). This terrane is bounded by the Tianshan–Mongolia Fold Belt to the north and separated from the Qaidam terrane by the Qilianshan Fold Belt to the south. The western boundary of the Alashan/Hexi Corridor terrane is not clearly delineated but is generally thought to lie in the vicinity of northeasterly extensions of the Altyn Tagh Fault beyond northern Qilianshan (Zhao *et al.* 1993).

Some geological data favour the view that the Alashan/Hexi Corridor terrane has been the western extension of the NCB since at least the Late Cambrian. First, the slightly metamorphosed pre-Cambrian basement of the Alashan/Hexi Corridor terrane is composed of the Lower Proterozoic Haiyuan Group in the Hexi Corridor area, the Lower Archaeozoic Helanshan Group in the northern part of the Helan Mountain, and the Lower Proterozoic Zhaochigou Group in Inner Mongolia along the Helan Mountain (NBGMR 1990). Like the other parts of the NCB, this pre-Cambrian metamorphosed basement and the Lower Cambrian deposits were mostly erased during continental uplift and denudation (Wu *et al.* 1993). Second, the most widely recognized regional unconformity in the stratigraphic sequence of the Alashan/Hexi Corridor terrane is between the Middle and Late Cambrian, which is also the case in the western Ordos Basin, suggesting that the Alashan/Hexi Corridor area might have belonged to the same tectonic unit as the Ordos Basin at that time (Zhao *et al.* 1993). Finally, the Alashan/Hexi Corridor area may have experienced the same continental uplift and denudation as the NCB during the mid-Palaeozoic, which correlates to the great stratigraphic hiatus in the period from the Upper Ordovician to the Lower Carboniferous in most of the NCB. In the Alashan/Hexi Corridor terrane, the upper Middle and Upper Ordovician series is absent, and the Silurian and Devonian rocks are mostly composed of epicontinental or continental sediments (NBGMR 1990). The disagreement in age of the stratigraphic hiatus is probably caused by the directional change of marine regression and transgression over the NCB (Yang *et al.* 1986). The unconformities between the Late Silurian to Middle Devonian and between the Late Devonian to Early Carboniferous indicate that folding occurred between these times in the Alashan/Hexi Corridor terrane (Fig. 1c; Zhao *et al.* 1993).

The Hexi Corridor, which is also called the Hexizoulang or Gansu Corridor, is located at the southern margin of the Alashan/Hexi Corridor terrane, and is bounded by two giant faults, the North Qilian fault and Longshoushan fault, running WNW–ESE on both the southern and northern edges. The Early Palaeozoic deposits are composed of the Middle Cambrian Xiangshan Group and the Lower–Middle Ordovician series in the eastern part of the Hexi Corridor. They are usually in contact with both the overlying and underlying strata with unconformity or disconformity (NBGMR 1990). The Silurian system in the eastern part of the Hexi Corridor is composed of the Lower–Middle Silurian Zhaohuajing Group and the Upper Silurian Hanxia Group, consisting largely of red beds with the exception of a few limestones and marlaceous limestones in the middle part of the Zhaohuajing Group. The Silurian groups provide some critical Silurian fossils, such as corals (*Pseudophaulactis*, *Tungussophyllum*, *Palaeofavosites*, *Mesofavosites*), brachiopods (*Holorhynchus*, *Eospirifer sinensis dasifiliformis*, *Clintonella*), and trilobites (*Coronocephalus*), assigning their palaeobiological age (NBGMR 1990). The continental Devonian rocks in the eastern part of the Hexi Corridor are divided into the Middle Devonian Shixiagou Formation and the Upper Devonian Zhongning Formation, whereas the Lower series is absent. Both the Shixiagou and Zhongning formations consist of red beds intercalated with a few limestones and marlaceous sandstones. The ages of these sediments are well controlled, with an abundance of fish remains of *Bothriolepis*, *Quasipetalichthys* (the Middle Devonian), and *Remigolepis* (the Upper Devonian), and flora of *Leptophloeum rhombicum*, *Sublepidodendron mirabile* (NBGMR 1990).

The Silurian strata unconformably overlie the Middle Cambrian Xiangshan Group and underlie the Eocene Sikouzi Formation. They are in contact with each other with paraconformity in the Zhaohuajing section. The Middle Devonian Shixiagou Formation unconformably overlies the Middle Cambrian Xiangshan Group and is unconformably overlain by the Upper Zhongning Formation, which paraconformably underlies the Middle Carboniferous Jingyuan Formation in the Shixiagou section and unconformably underlies the Early Carboniferous Chouniugou Formation in the Xiaoyuchuan section (Figs 1b and c).

For the Silurian, 37 orientated drill-core limestone samples from three sites and 20 orientated drill-core red bed samples from two sites were collected from the Early–Middle Silurian Zhaohuajing Group, Zhaohuajing section. In addition, 35 orientated drill-core red bed samples from four sites were collected from the Upper Silurian Hanxia Group, Zhaohuajing section. For the Devonian, 69 orientated drill-core red bed samples from seven sites were collected from the Middle Devonian Shixiagou Formation, Shixiagou section. 23 orientated drill-core red bed samples from two sites, 25 orientated drill-core marlaceous sandstone samples from two sites, and 10 orientated drill-core limestone samples from one site were all

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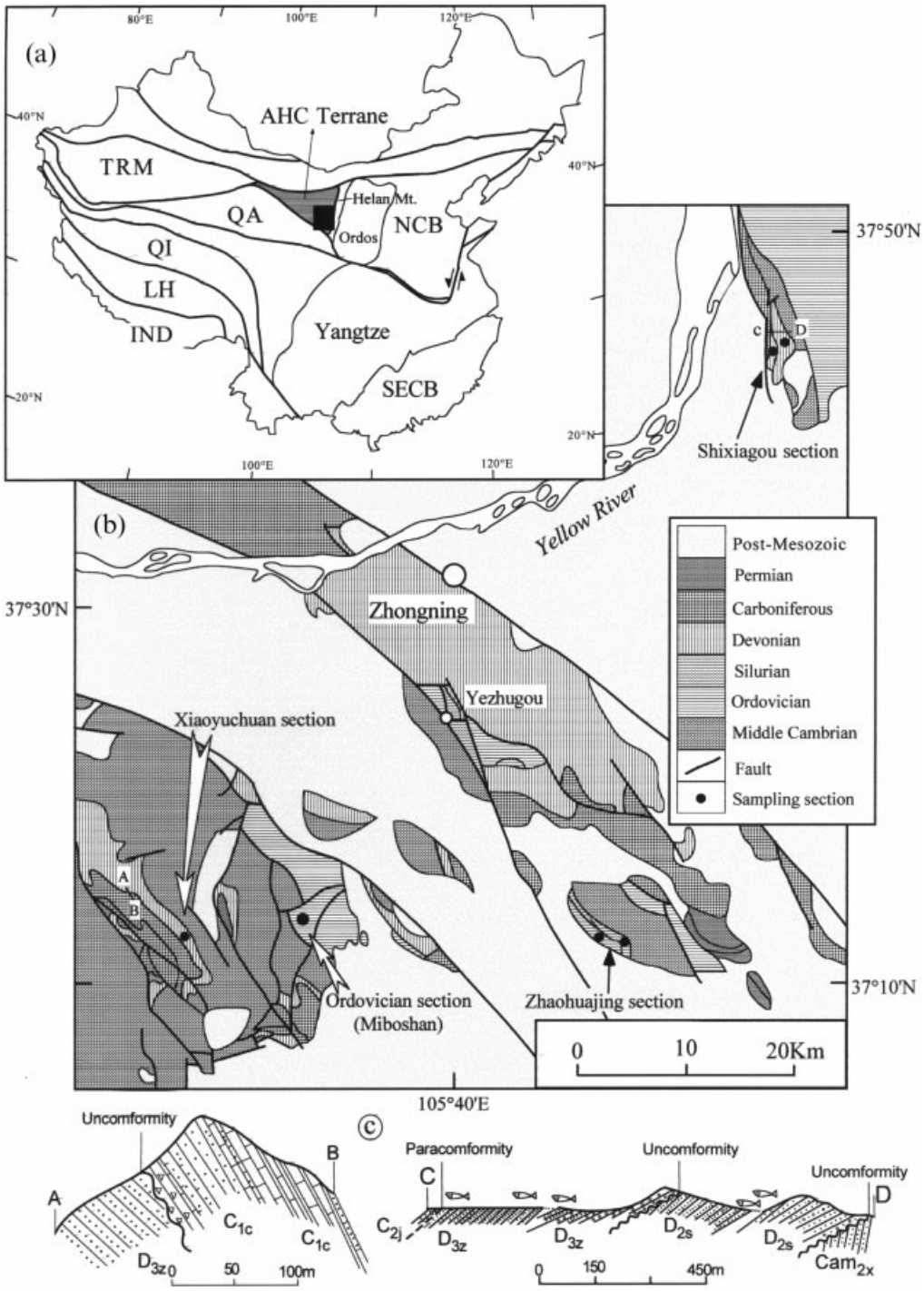


Figure 1. Sketch maps of (a) the present locations of the continental blocks of China showing the tectonic framework and (b) Zhongning area, southwest of Ningxia Hui Autonomous Region showing the Palaeozoic geological stratigraphy and the distribution of palaeomagnetic sampling sections, and (c) two cross-sections in the sampling areas showing the unconformity of the Devonian with younger strata. AHC terrane: Alashan/Hexi Corridor terrane; IND: India block; LH: Lhasa (South Tibet) block; NCB: North China block; QA: Qaidam block; QI: Qiangtang (North Tibet) block; SECB: southeast China block; TAR: Tarim block; Yangtze: Yangtze block; C_{2j}: Middle Carboniferous Jingyuan Formation; C_{1c}: Early Carboniferous Chouniugou Formation; D_{3z}: Late Devonian Zhongning Formation; D_{2s}: Middle Devonian Shixiagou Formation; and Cam_{2x}: Middle Cambrian Xiangshan Group.

collected from the Upper Devonian Zhongning Formation, Shixiagou section, and 20 orientated drill-core red bed samples from two sites were collected from the Upper Devonian Zhongning Formation of the Xiaoyuchuan section (Fig. 1b and Tables 1, 2). Each site consists of about 10 independently orientated drill-core samples covering several metres or more. All the samples were collected with a portable drill and orientated using a magnetic compass.

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Table 1. Summary of the Silurian and Devonian results of the A component from the Hexi Corridor terrane.

Site ID	Age	Strike/dip	R/N	n/n_0	Range (°C)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k_s	α_{95}	G_{Long}	G_{Lat}	S_{Long}	S_{Lat}
Silurian															
Zhaohuajing area (105.9°E, 37.2°N): limestones															
YS07	S1–2	140/30	0/3	3/4	250–530	39.8	55.3	356.1	82.8	65.2	15.4	186.1	58.2	104.4	51.3
YS08	S1–2	145/41	0/9	9/9	300–590	12.8	58.5	287.6	63.8	35.0	8.8	180.7	79.8	49.2	36.9
YS09	S1–2	145/41	0/11	11/11	100–530	20.8	54.5	297.5	68.4	52.9	6.3	197.0	73.1	55.2	44.6
Zhaohuajing area (105.9°E, 37.2°N): red beds															
YS10	S1–2	148/40	0/6	6/9	300–560	40.2	46.4	338.1	77.6	16.7	17.1	199.8	55.1	89.3	58.1
YS11	S1–2	140/45	2/3	5/5	590–680	292.2	64.7	256.1	30.7	14.2	21.1	49.5	40.2	37.4	−0.7
YS12	S3	138/58	0/4	4/10	100–400	312.5	59.4	261.8	24.5	41.3	14.5	35.7	53.3	31.0	1.3
YS13	S3	128/45	0/5	5/7	200–560	4.8	21.8	337.4	54.3	12.2	22.8	275.2	63.8	15.3	71.6
YS14	S3	128/45	0/7	7/7	300–530	14.5	40.6	314.1	72.3	14.9	16.2	240.2	71.3	64.9	53.9
YS15	S3	105/48.5	0/3	3/4	300–560	3.2	43.9	265.4	81.0	43.3	19.0	271.7	78.2	84.6	33.9
Middle and Late Devonian															
Shixiagou area (106.0°E, 37.7°N): red beds															
YD16	D2	216/27	0/4	4/5	300–560	70.6	52.5	30.0	59.8	27.6	17.8	177.7	33.6	179.3	66.7
YD17	D2	214/26	0/9	9/9	350–560	53.0	44.4	25.8	46.9	27.1	10.1	194.6	44.2	212.4	66.4
YD18	D2	214/25	0/6	6/9	400–560	86.2	68.6	3.5	74.9	10.0	22.3	151.8	30.9	110.1	66.0
YD19	D2	217/27	0/11	11/11	300–560	66.4	50.2	29.9	55.8	11.0	14.4	181.9	35.9	190.0	66.2
YD20	D2	214/23	0/7	7/11	350–590	24.6	73.8	337.3	60.0	23.7	12.7	133.3	62.9	33.4	72.2
YD23	D3	173/21	0/8	8/9	300–590	57.1	58.2	23.9	74.4	19.5	12.9	176.0	45.9	131.3	62.4
YD24	D3	183/23	0/7	7/10	300–560	30.2	58.9	348.7	61.7	12.6	17.7	181.8	66.5	50.6	80.0
Shixiagou area (106.0°E, 37.7°N): marlaceous sandstones															
YD25	D3182/23		0/8	8/8	350–590	287.6	52.7	282.9	30.2	153.4	4.5	35.3	32.4	21.5	19.9
YD26	D3182/23		0/7	7/10	350–590	60.9	75.3	308.4	77.2	22.1	13.1	141.8	46.1	76.1	49.5
Shixiagou area (106.0°E, 37.7°N): limestones															
YD27	D3183/23		0/5	5/9	20–60 mT	116.1	47.5	136.7	66.9	220.9	5.21	58.0	−0.7	132.5	5.3
Xiaoyuchuan area (105.4°E, 37.2°N): red beds															
YD38	D3152/31		1/1	2/5	200–500	252.8	20.6	252.3	−9.9	–	–	34.4	−6.9	22.2–17.1	
YD39	D3151/28		0/4	4/6	300–560	359.1	48.7	323.2	54.1	15.2	37.6	291.3	82.4	21.0	60.3
Averaging by site				#$N = 21$		31.6	59.8	330.8	72.1	6.3	13.8	172.5	65.9	76.0	62.3

Abbreviations: Site ID, site identification; Strike/dip, strike azimuth and dip of bed; R/N, samples show reversed and normal polarity; n/n_0, number of samples in the calculation/number of samples demagnetized; Range, demagnetization interval; D_g, I_g (D_s, I_s), declination and inclination of direction *in situ* (after tilt-correction); G_{Long}, G_{Lat} (S_{Long}, S_{Lat}), longitude and latitude of VGP in geographical (stratigraphic) coordinates. S, D, Silurian and Devonian with 1 = Early, 2 = Middle, 3 = Late. Coordinate system: geographical coordinates (g), stratigraphic coordinates (s). # The site mean direction was calculated after the directions were transferred to the sampling site of 106.0°E, 37.7°N, Shixiagou area.

3 ROCK MAGNETIC STUDY

3.1 Techniques

Acquisition of isothermal remanent magnetization (IRM) and progressive thermal demagnetization of composite IRMs were carried out to identify coercivity and unblocking temperature spectra for 29 representative samples, one or two from each site. A direct current (DC) magnetic field was applied for the acquisition of IRM with a 2G Enterprises Pulse Magnetizer with a maximum field of up to 2700 mT. The IRM was measured on a spinner magnetometer (Natsuhara SMM-85). These samples were also subjected to progressive thermal demagnetization at temperatures of 100, 200, 300 °C, and then in intervals of 50, 30 or 20 °C to 690 °C after applying a composite IRM following Lowrie (1990). DC magnetic fields of 2700, 400, and 140 mT were applied to the sample *Z*-axis, *Y*-axis and *X*-axis, respectively. Remanent magnetizations were measured with a 2G Enterprises cryogenic magnetometer at Kobe University in Japan.

A few pilot red bed samples were also subjected to low-temperature and thermomagnetic analysis experiments in order to identify low-temperature magnetic phase transitions and Curie temperatures of the samples. The low-temperature experiments were carried out on a MPMS superconducting susceptometer at Kyoto University. A high DC magnetic field (1.0 T) was imparted to the sample at 5 K and the IRM was subsequently measured while warming the sample to room temperature. In the thermomagnetic analysis, samples were heated up to 700 °C at a constant rate of 8 °C per minute and cooled to room temperature at the same rate in an argon gas atmosphere.

3.2 Results

Rock magnetic experiments imply the ferromagnetic predominance of haematite and subordinate magnetite for the red bed samples (Fig. 2). The acquisition curve of IRM shows samples reaching quasi-saturation at about 1500 mT in a DC

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Table 2. Summary of the Silurian and Devonian palaeomagnetic sampling and results of the characteristic B component from the Hexi Corridor terrane.

Site ID	Age	Strike/dip	R/N	n/n_0	Range (°C)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k_s	α_{95}	G_{Long}	G_{Lat}	S_{Long}	S_{Lat}
Silurian															
Zhaohuajing area (105.9°E, 37.2°N): limestones															
YS07	S1–2	140/30	0/2	2/4	400–590	354.6	22.4	339.1	36.4	–	–	298.0	64.0	338.7	65.1
YS08	S1–2	145/41	0/2	2/9	530–590	342.2	24.1	320.8	29.1	–	–	323.5	60.7	354.6	49.2
Zhaohuajing area (105.9°E, 37.2°N): red beds															
YS10	S1–2	148/40	2/6	8/9	620–680	344.8	20.7	326.8	26.4	10.5	17.9	317.1	60.2	346.5	52.4
YS12	S3	138/58	0/10	10/10	590–680	12.9	−8.6	2.4	37.3	25.6	9.7	266.9	46.8	278.0	73.5
YS13	S3	128/45	0/6	6/7	610–690	340.4	−6.1	336.3	17.6	16.8	16.8	314.6	45.8	329.2	54.6
YS14	S3	128/45	0/7	7/7	610–690	355.7	1.8	344.7	33.0	21.8	13.2	293.1	53.5	325.0	66.6
YS15	S3	105/48.5	1/3	4/4	620–680	327.9	−5.5	320.5	26.3	8.1	34.4	329.9	40.2	352.9	47.9
Middle and Late Devonian															
Shixiagou area (106.0°E, 37.7°N): red beds															
YD16	D2	216/27	1/0	4(3)/5	650–690	179.5	−37.9	168.3	−19.5	21.5	27.5	107.6	−73.6	129.8	−60.4
YD17	D2	214/26	0/7	7/9	620–690	35.2	48.0	357.0	32.6	42.4	8.82	02.0	59.6	294.3	69.9
YD18	D2	214/25	4/2	6/9	620–690	140.7	−56.7	134.7	−32.5	10.8	21.3	207.5	−59.0	181.9	−45.7
YD19	D2	217/27	1/0	10(9)/11	620–690	152.8	−31.6	148.9	−6.8	17.5	13.9	162.7	−58.5	153.3	−45.5
YD20	D2	214/23	0/5	7(2)/11	620–660	9.6	35.8	358.0	24.1	12.4	18.3	258.3	70.3	290.6	64.8
YD21	D2	208/24	0/9	9/9	400–670	336.6	49.3	325.8	29.1	27.1	10.1	1.7	69.2	349.1	52.6
YD22	D2	209/24	0/5	5/5	450–690	335.7	38.0	328.6	17.7	178.5	5.03	44.9	63.4	338.8	49.8
YD23	D3	173/21	8/0	8/9	620–690	153.8	−26.7	145.7	−18.3	8.6	20.1	157.6	−56.9	162.4	−48.1
YD24	D3	183/23	1/3	7(3)/10	590–680	346.3	38.6	332.0	29.1	13.0	18.0	326.4	70.2	341.7	56.8
Shixiagou area (106.0°E, 37.7°N): marlaceous sandstones															
YD26	D3	182/23	1/2	6(3)/10	400–660	337.6	41.6	323.3	29.4	39.5	11.6	346.7	66.5	351.9	50.9
Xiaoyuchuan area (105.4°E, 37.2°N): red beds															
YD38	D3	152/31	2/0	2/5	560–690	141.7	−26.9	129.1	−17.8	–	–	171.7	−49.0	177.3	−36.3
YD39	D3	151/28	1/2	3/6	590–690	4.1	9.9	356.5	23.9	25.7	24.0	277.8	57.6	293.5	65.1

Abbreviations: n/n_0, number of samples in the calculation (number of remagnetization circles used)/number of samples demagnetized. The rest of the abbreviations as in Table 1.

field (Fig. 2a), indicative of the presence of a high-coercivity magnetic mineral. Thermal demagnetization of the composite IRMs shows that sample remanence is mainly composed of hard to medium components with unblocking temperatures of about 690 °C (Fig. 2b), indicating the predominance of haematite. The Curie temperature of haematite is also detected by thermomagnetic analysis (Fig. 2c). A few samples show magnetite in the sample because of significant unblocking around 550 °C of composite IRMs, and the conspicuous magnetic phase transition at about 120 K (Fig. 2d), which suggests that magnetite is present with larger grain sizes such as pseudo-single-domain (PSD) or multidomain (MD, see Halgedahl & Jarrard 1995).

Haematite and magnetite are also expected to be predominant in the marlaceous sandstones. The acquisition behaviour of IRM shows a trend of quasi-saturation in relatively low fields and consequently yields quasi-saturation at about 1500 mT (Fig. 3a, sample YD258). Progressive thermal demagnetization of the composite IRMs reveals that the hard component is unblocked by about 680 °C, and that the medium and soft components are unblocked by about 580 °C (Fig. 3b).

The IRM acquisition experiments and thermal demagnetization of composite IRMs show that the limestone samples are dominated by magnetite. Indeed, the IRM of the sample usually reaches quasi-saturation at lower than 200 mT (Fig. 3a, sample YS99), and the thermal demagnetization of the soft, medium, and hard components shows only one unblocking temperature of about 580 °C (Fig. 3c).

4 PALAEOMAGNETIC STUDY

4.1 Techniques

In the laboratory, with the exception of a few limestone samples which were subjected to hybrid thermal and AF demagnetization owing to their unstable nature at higher temperatures, all the samples were progressively thermally demagnetized with a thermal demagnetizer (Natsuhara TSD-1). Demagnetization intervals were as large as 100 °C at the lower temperatures and as small as 10 °C at the higher temperatures. The hybrid demagnetization comprised progressive thermal demagnetization up to 300 °C and subsequent AF demagnetization with an alternating current (AC) field interval of 5 mT for the lower field and 20 mT for the higher field. Progressive AF demagnetization was executed on an AF demagnetizer (Natsuhara DEM-8601C).

Demagnetization results were plotted on orthogonal diagrams (Zijderveld 1967) and stereographic projections. Magnetic components were separated using principal component analysis (Kirschvink 1980) and calculated using Fisher statistics (Fisher 1953) or the method of McFadden & McElhinny (1988) when remagnetization circles were used.

4.2 Demagnetization results

Demagnetization results for 163 out of the 239 samples collected yield relatively simple demagnetization behaviours

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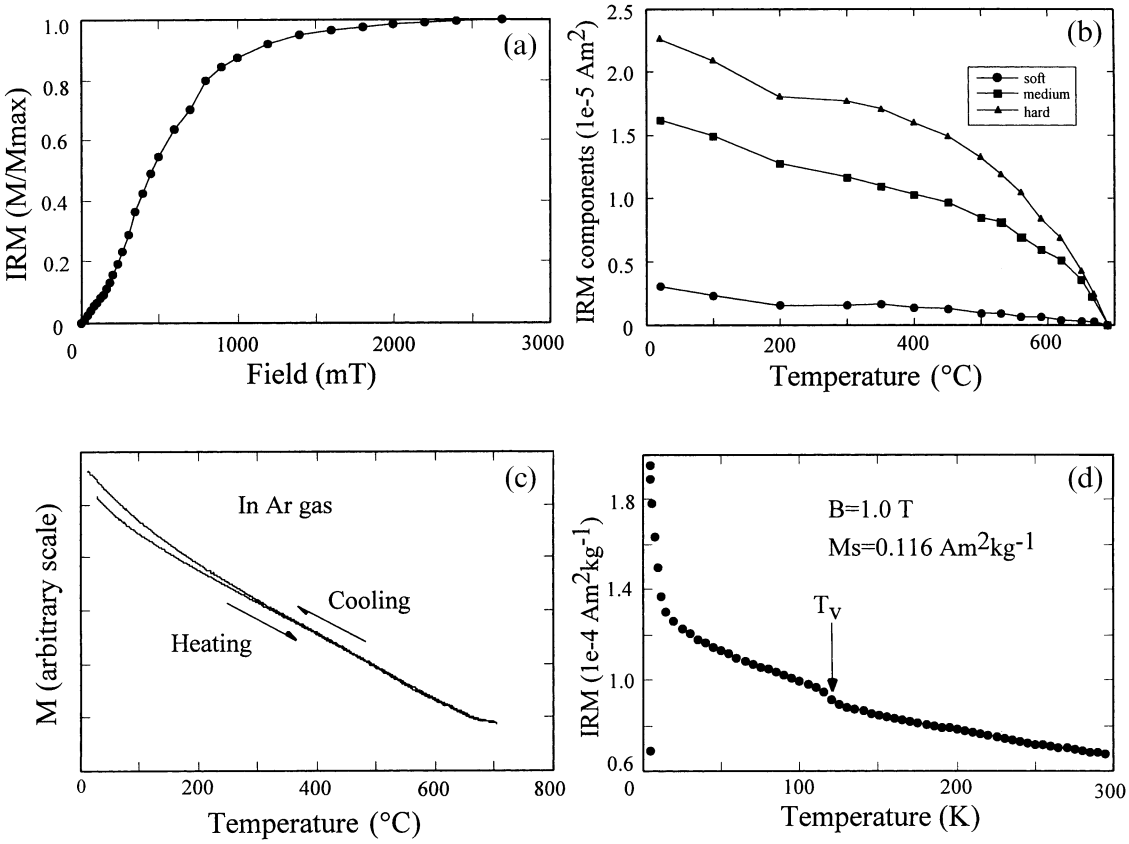


Figure 2. Rock magnetic behaviour of red bed sample YS133: (a) acquisition of isothermal remanent magnetization (IRM); (b) progressive thermal demagnetization curves of composite IRMs following Lowrie (1990); (c) Curie temperature measurement showing the change of IRM with heating and cooling in an argon gas atmosphere; and (d) low-temperature magnetic phase transitions revealed by the warming up of IRM imparted at 5 K with a DC field of 1.0 T.

for both Silurian and Devonian samples after the removal of recent viscous overprint components by temperature steps of 100 or 200 °C (Figs 4, 5 and 6).

4.2.1 Silurian red beds

The thermal demagnetization of Silurian red bed samples is characterized by the appearance of middle- and high-temperature components with different directions (Fig. 4). The middle-temperature component, designated component A hereafter, was generally removed by about 560 °C. This component is mostly of normal polarity with NNE declination and steeply downward inclination *in situ*, except for a few samples from site YS12 which has a westerly declination (Fig. 4c). This middle-temperature component A can be removed by about 250–400 °C or so in a few cases (Fig. 4c). The high-temperature component, designated component B, was usually unblocked by 680 or 690 °C. This component is mostly of normal polarity and reveals NNW and shallow downward directions *in situ*. An anomalous direction of this high-temperature component was observed in a few samples at site YS11, which is directed to the SW–NW and has a steep downward direction *in situ*, and is subparallel to the direction of the middle-temperature component A (Table 1 and Fig. 4d).

4.2.2 Silurian limestones (sites YS7, YS8 and YS9)

Thermal demagnetization for Silurian limestone samples indicates a stable remanence up to 585 °C, with vector end-points trending into the origin (Fig. 5). Most samples preserve a stable component that parallels the component A of the red bed samples and has a NE direction with moderate to steep downward inclination *in situ* (Fig. 5a). A few samples yield an additional component that clusters around directions with northerly declination and shallow downward inclination *in situ* (Fig. 5b). The directions of these samples parallel those of the B component of the red bed samples.

4.2.3 Devonian red beds and marlaceous sandstones

The demagnetization results of Devonian red bed and marlaceous sandstone samples are divided into two groups. First, most samples exhibit demagnetization behaviour similar to that of Silurian red bed samples, for example soft viscous overprints, middle-temperature component A and high-temperature component B (Figs 6a and c). Some samples show a low-temperature component between demagnetization temperatures of 100 and 300 °C, and the *in situ* direction is similar to that of the present field (Figs 6c and d). The high-temperature component B, however, cannot be well isolated in some samples even when the demagnetization temperature is increased to about 650 °C (Fig. 6d), in which the middle-temperature component A almost obliterates the high-temperature characteristic remanent magnetization. In these cases, site-mean observation of the B component is calculated by combining remagnetization great circles with direction observations using McFadden & McElhinny's (1988) method.

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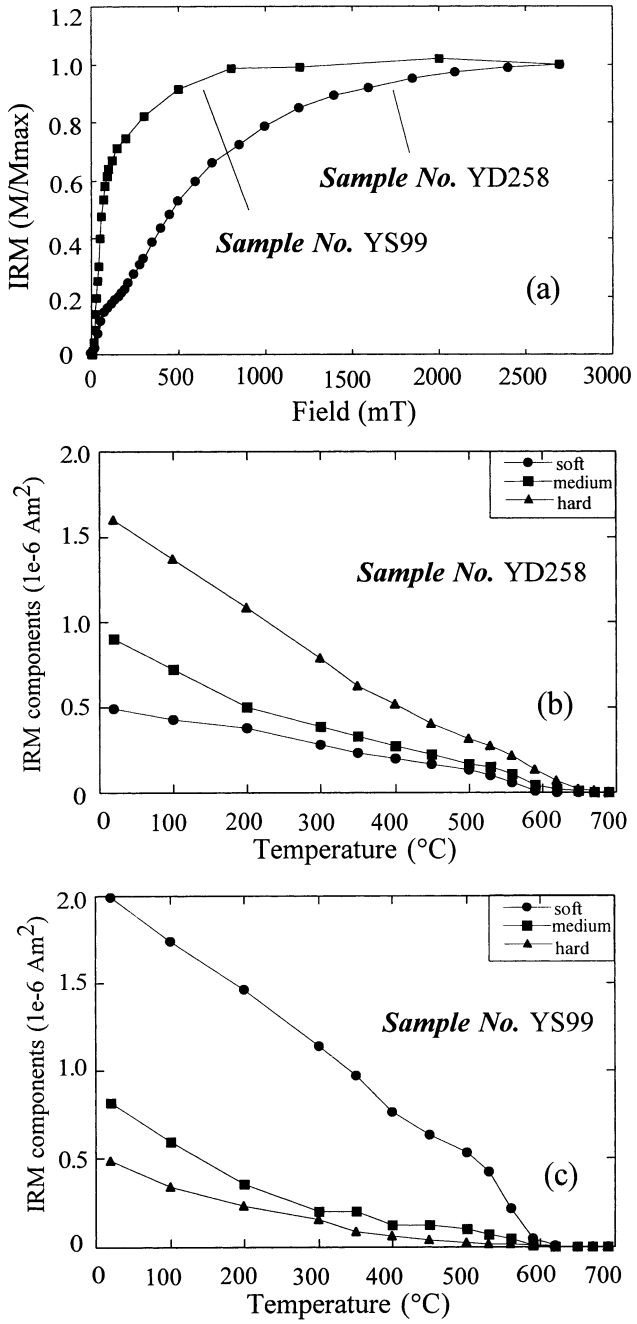


Figure 3. Behaviour of the acquisition of IRM and progressive thermal demagnetization curves of composite IRMs following Lowrie (1990).

Second, the thermal demagnetization behaviour of samples from the two sites YD21 and YD22 is much simpler (Fig. 6b). After the removal of soft viscous overprints in the initial stages of cleaning, demagnetization trajectories successively trend into the origin, indicating one stable component that is demagnetized before the unblocking. This stable component was unblocked by about 680 °C, and is designated component B. Demagnetization results for marlaceous sandstone samples from site YD25 exhibit only the middle-temperature component A by about 590 °C (Fig. 6e).

4.2.4 Devonian limestones (site YD27)

The demagnetization results for Devonian limestone samples are unstable and no characteristic component was isolated when progressive thermal demagnetization was applied. The hybrid demagnetization technique isolates one stable remanence from 20 mT to 60 or 80 mT (Fig. 6f); however, it is insignificantly different in direction from the middle-temperature component A of the Devonian red bed and marlaceous sandstone samples.

The site-mean directions, corresponding VGP positions, and associated statistical parameters for the components A and B are listed in Tables 1 and 2, respectively.

4.3 Field test for stability

The component A is directed to the NNE with about 50° or 60° downward inclination *in situ*. This component is usually removed by about 590 °C, although it can obliterate the high-temperature characteristic component B up to about 680 °C in a few cases. It is dominated by normal polarity and has an *in situ* mean direction of $D = 31.6°$, $I = 59.8°$, $\kappa = 6.6$, $\alpha_{95} = 13.4°$ for all the Silurian and Devonian samples (Fig. 7). The fold test is inconclusive by the traditional method (McElhinny 1964). Nonetheless, the fold test result using McFadden's (1990) method (*in situ* $\xi_2 = 6.955$, tilt-corrected $\xi_2 = 7.635$, $\xi = 7.483$, critical value at 99 per cent confidence) enables us to say that there is no case for rejecting the hypothesis that this component postdates the age of folding.

The component B can be isolated from Silurian and Devonian red beds, marlaceous sandstones, and a few limestones in three sampling localities. This component has an *in situ* mean direction of $D = 348.3°$, $I = 7.1°$, $\kappa = 15.8$, $\alpha_{95} = 15.7°$ and a tilt-corrected one of $D = 335.2°$, $I = 30.1$, $\kappa = 31.8$, $\alpha_{95} = 10.9°$ for the Silurian samples. For the Devonian samples, it has an *in situ* mean direction of $D = 345.9°$, $I = 38.6°$, $\kappa = 15.8$, $\alpha_{95} = 11.3°$ and a tilt-corrected one of $D = 333.9°$, $I = 24.3$, $\kappa = 22.7$, $\alpha_{95} = 9.3°$ (Table 3). The fold test is positive at the 99 per cent confidence level for the Silurian and Devonian samples by McElhinny's (1964) fold test ($\kappa_s/\kappa_g = 2.382 > 2.206$, critical value at the 99 per cent confidence level). This indicates that the characteristic B component has a prefolding origin and was acquired prior to the Early Carboniferous. The B component is mostly of normal polarity for the Silurian samples and is of both normal and reversed polarity for the Devonian ones (Fig. 8). The rock magnetic experiments and the demagnetization characters suggest that this component probably resides in haematite in the red bed and marlaceous sandstone samples, and in magnetite in the limestone samples. A consistency analysis (McFadden & Jones 1981) of the normal and reversed directions for the Devonian shows that they share the same component at the 95 per cent confidence level ($F = 2.72 < 3.49$, critical value at the 95 per cent confidence level), indicative of a positive reversal test. Application of McFadden & McElhinny's (1990) classification of the reversal test shows a class C reversal test ($\gamma = 15.6° < 17.7°$, critical value at the 95 per cent confidence level).

5 DISCUSSION

The fold test demonstrates that component A was probably acquired after folding while the characteristic component B was acquired before folding. The corresponding *in situ* palaeomagnetic

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Table 3. Grand mean of normal and reversed characteristic directions.

P	*n*/*N*	C	Direction				Pole position			
			D	*I*	α_{95}	κ	Long.	Lat.	A_{95}	*K*
Silurian (105.9°E, 37.2°N)										
N	7/7	G	348	37.1	15.7	15.8	306.5	54.9	12.2	25.3
		S	335.2	30.1	10.9	31.8	339.0	60.1	11.2	30.1
Middle and Late Devonian (106.0°E, 37.7°N)										
N	7/12	G	354.9	39.3	16.0	15.2	304.7	74.6	16.7	14.0
		S	340.2	27.5	11.5	28.6	328.6	61.0	12.0	26.1
R	5/12	G	154.1	−36.7	16.7	21.8	167.4	−62.2	16.7	22.1
		S	145.	−19.4	16.2	23.4	163.5	−48.5	15.0	27.0
M	12/12	G	345.9	38.6	11.3	15.8	328.9	70.6	11.9	14.3
		S	333.9	24.3	9.3	22.7	336.0	56.0	9.2	23.1

Abbreviations: P = polarity: N = normal; R = reversed; M = both N and R; *n*/*N* = number of sites used in statistical analysis/total of sites exhibiting the characteristic B component; C = coordinates: G = Geographic; S = Stratigraphic. The mean directions of the Middle and Late Devonian were calculated after the site observations were transferred to the sampling site of 105.9°E, 37.2°N. The rest are as in Table 1.

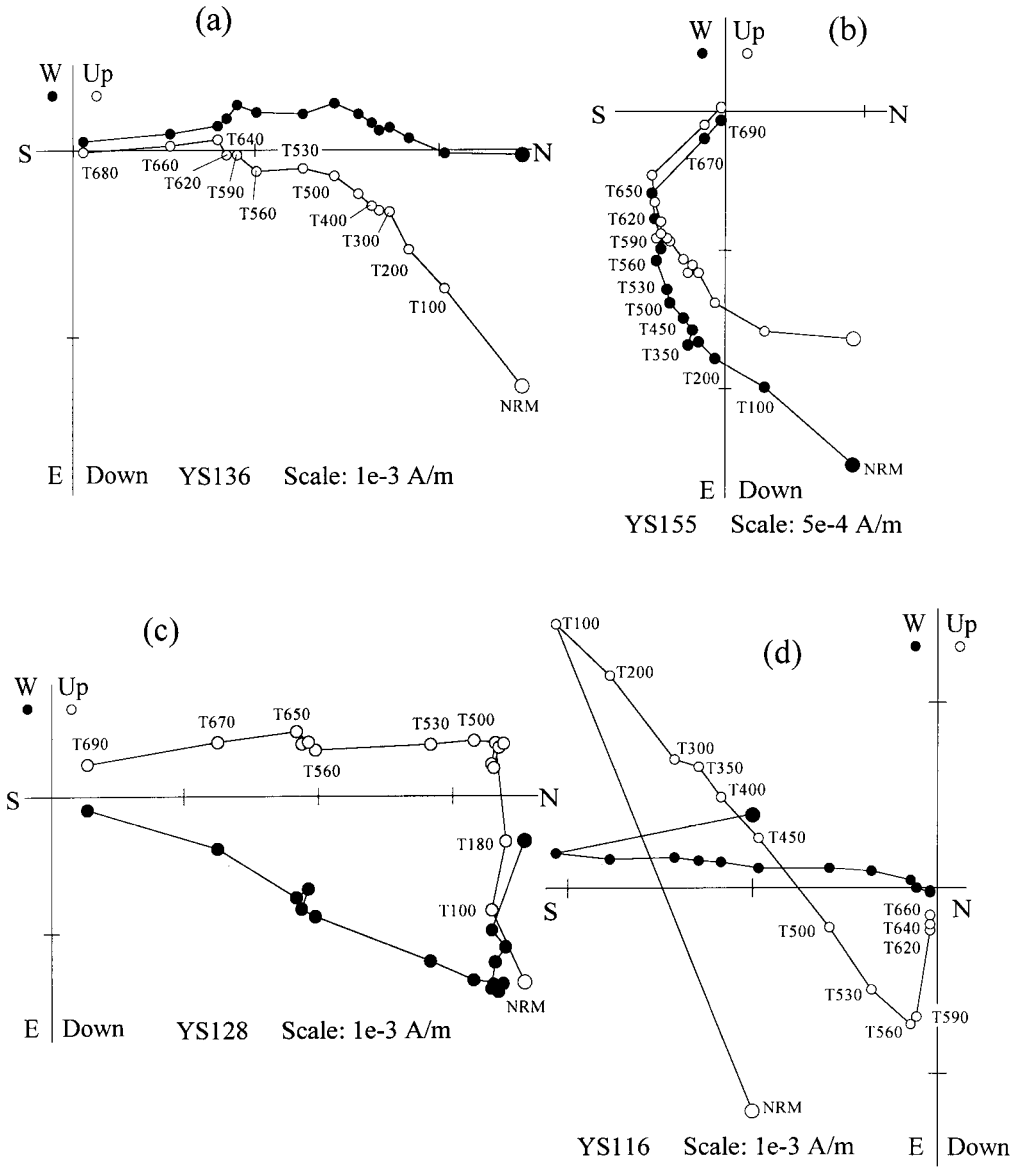


Figure 4. Orthogonal vector plots of representative Silurian red bed samples. Directions are plotted *in situ*; open and solid circles represent vector endpoints projected onto the vertical and horizontal planes, respectively. T indicates thermal (°C) demagnetization.

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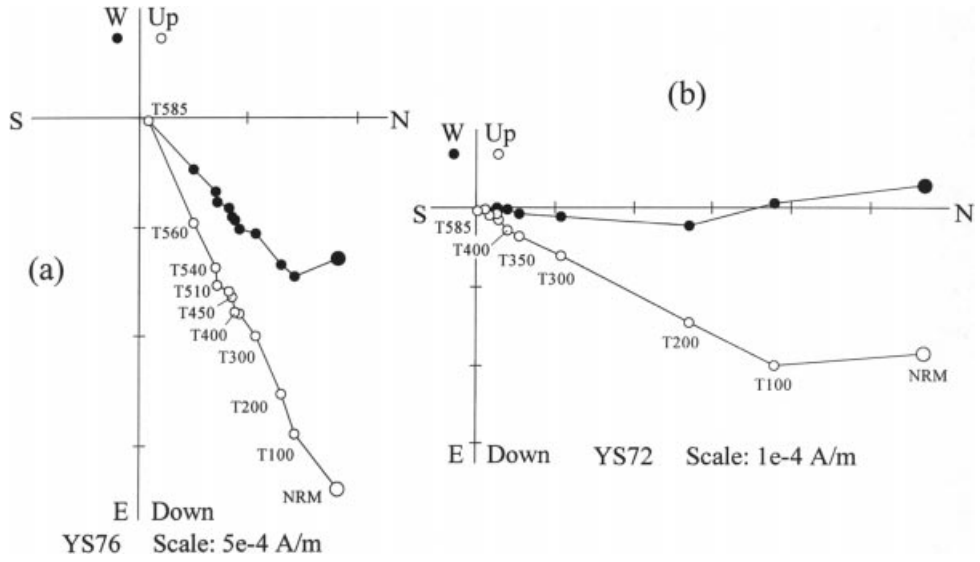


Figure 5. Orthogonal vector plots of representative Silurian limestone samples. Symbols as for Fig. 4.

pole of component A is at 172.5°E, 65.9°N with $K = 4.5$, $A_{95} = 16.9°$ and is located close to poles of Middle Jurassic to Late Cretaceous age for the NCB (Zhao *et al.* 1994) at the 95 per cent confidence limit. This ubiquitous component A is therefore probably a Middle Jurassic to Late Cretaceous overprint.

The prefolding nature of component B is also ascertained by McFadden's (1990) fold test. Indeed, the fold test is positive at the 99 per cent confidence level for the Silurian and Devonian individual VGPs (geographical $\xi_1 = 8.434$, stratigraphic $\xi_1 = 1.445$, $\xi = 7.112$, critical value at 99 per cent confidence). The rock magnetic experiments and the demagnetization characters suggest that this component resides in haematite in the red bed and marlaceous sandstone samples, and in magnetite in the limestone samples. The positive reversal test and the pervasive existence in rocks of different lithologies and widely distributed localities imply that the characteristic high-temperature component B was acquired in, or close to, the period when the Silurian–Devonian rocks formed. Two palaeomagnetic poles are therefore calculated for the Silurian–Devonian period (Table 3).

5.1 Comparison with previous results

The newly obtained Silurian and Devonian palaeomagnetic poles are at 339.0°E, 60.1°N with $A_{95} = 11.2°$, and 336.0°E, 56.0°N with $A_{95} = 9.2°$, respectively. Previously, a few Silurian and Devonian palaeomagnetic results reported from the Alashan/Hexi Corridor terrane showed a high level of contradiction among themselves (Fig. 9).

The Silurian and Devonian palaeomagnetic poles of this study show good consistency with the Late Silurian pole of Li *et al.* (1985). The Devonian pole of Zhu & He (1987) is also located close to our poles (Fig. 9). These palaeopoles are located a significant distance from the other Silurian and Devonian palaeopoles, as well as from the Late Palaeozoic palaeopole from the same terrane (Meng *et al.* 1990). The Devonian pole of Zhu & He (1987) is calculated from 97 sandstone and limestone samples collected from the northern margin of the Alashan/Hexi Corridor terrane and spanning the interval from the Early to Late Devonian. Because this study used endpoint analysis without principal component analysis, however, this palaeopole is questionable. The Late Silurian pole reported by Li *et al.* (1985) is from the middle part of the Hexi Corridor, Gansu Province; it passes a fold test but not much detail is presented about the origin. The consistency between our Silurian and Devonian palaeomagnetic poles and that of Li *et al.* (1985) may exclude relative rotation between the eastern and middle parts of the Hexi Corridor terrane.

Another Silurian and Devonian palaeomagnetic pole group consists of the palaeopoles of He *et al.* (1988) and Zhao *et al.* (1993). These palaeopoles are located conspicuously far away from both the first Silurian and Devonian palaeopole clusters and the Late Palaeozoic pole from the same terrane (Fig. 9). Among these palaeopoles, the Silurian one presented by He *et al.* (1988) was calculated from only nine red bed samples collected from near Zhongning County. Because of the incomplete thermal demagnetization, only up to 600 °C, and the absence of principal component analysis with the pole of Zhu & He (1987), this palaeopole is possibly a remagnetization and is regarded as unreliable. The Silurian and Devonian palaeopoles presented by Zhao *et al.* (1993) are located at 228.4°E, 26.2°N with $A_{95} = 8.2°$ for the Silurian, and 228.7°E, 34.2°N with $A_{95} = 8.8°$ for the Devonian. These palaeopoles are calculated from the high-temperature component isolated from red beds, andesitic basalt, grey sandstone, and a few limestone rocks at widely distributed localities and pass fold and reversal tests.

Two other Silurian and Devonian palaeomagnetic poles are reported from the Late Devonian Zhongning Formation in the eastern part of the Hexi Corridor, Ningxia (He *et al.* 1988), and the Late Silurian Hanxia Formation in the western part of the Hexi Corridor, Gansu Province (Meng & Coe 1992). The provisional palaeomagnetic pole of the former study (349.6°E, 4.4°N, δp, $\delta m = 21.1°$, 37.4°) is obviously unreliable because it was calculated from only five red bed samples with incomplete thermal demagnetization, up to 600 °C. The latter palaeomagnetic pole (5.5°E, 45.1°N, $A_{95} = 4.7°$) was determined from 38 purplish red sandstone and mudstone samples, and passes the fold test. The small deviation between this Late

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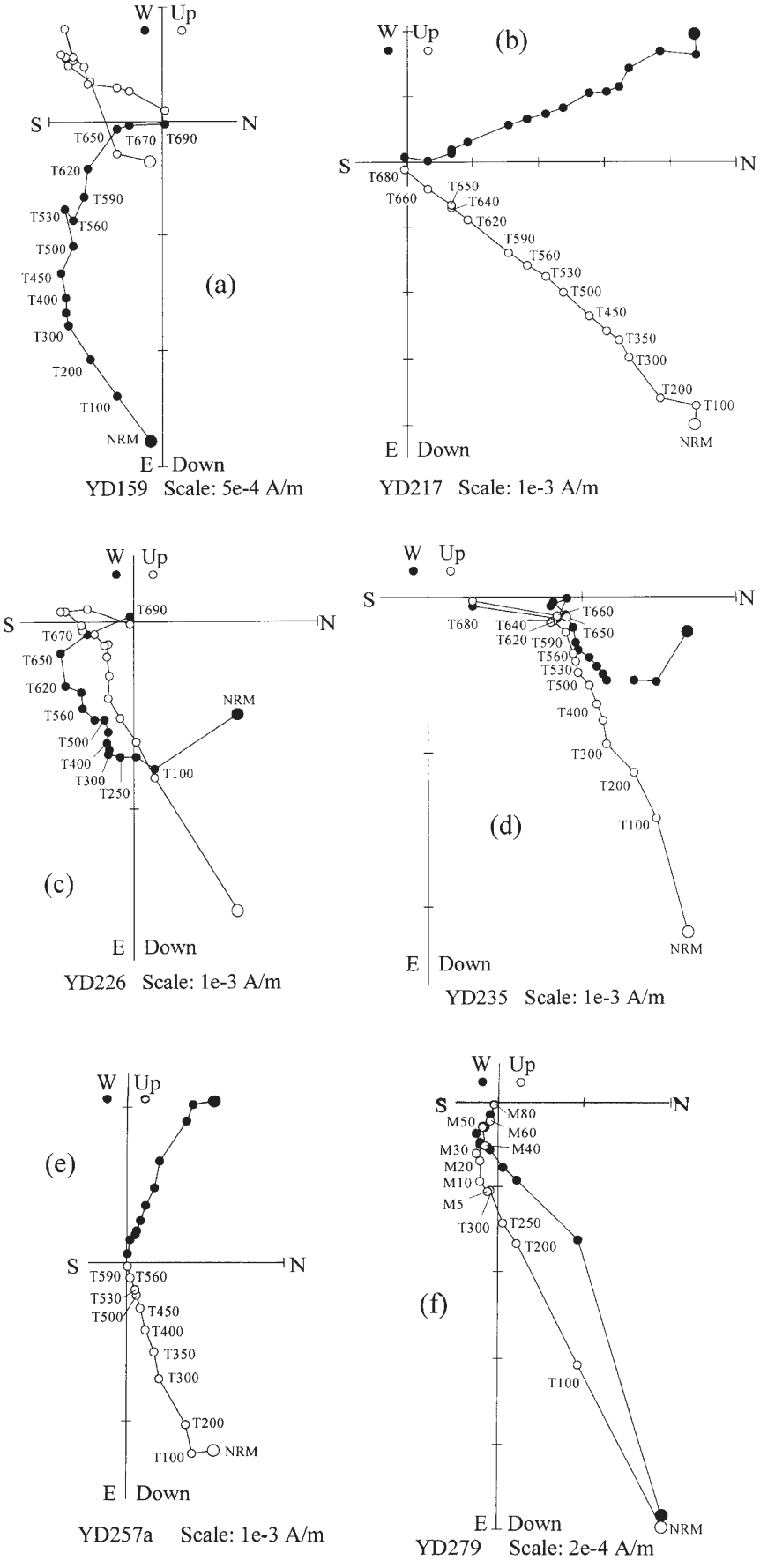


Figure 6. Orthogonal vector plots of representative Devonian samples. M indicates AF (mT) demagnetization. Symbols as for Fig. 4.

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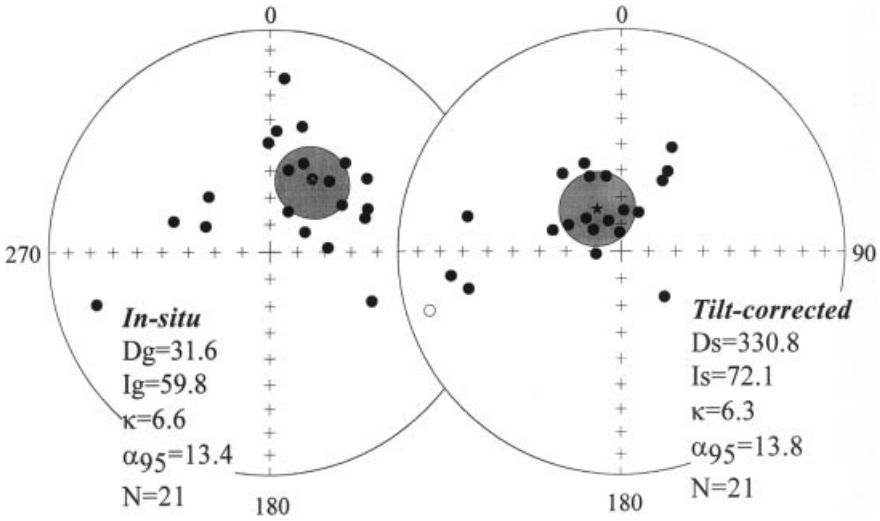


Figure 7. Equal-area projection of component A from the Silurian and Devonian samples before and after tilt correction. Solid/open symbols represent downward/upward dipping directions. The star is the mean direction shown with the 95 per cent confidence limit.

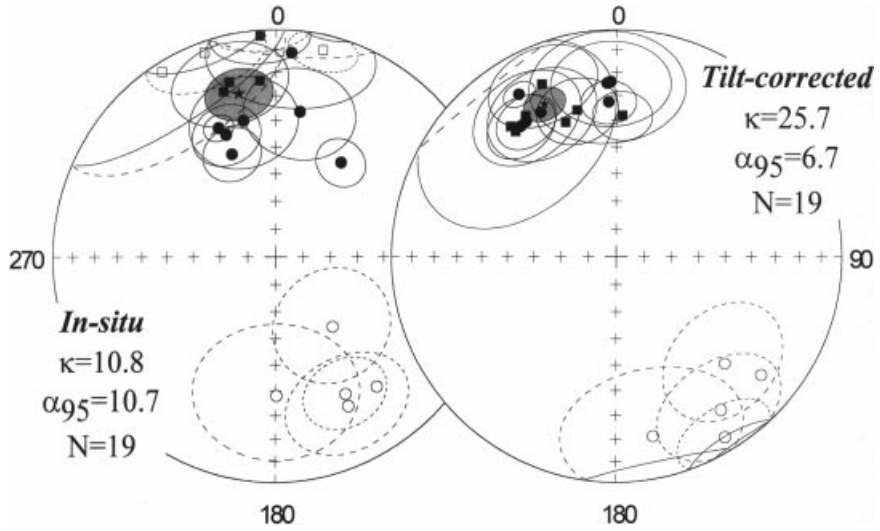


Figure 8. Equal-area projection of the Silurian (squares) and Devonian (circles) site-mean characteristic remanent magnetization (component B) directions before and after tilt correction. Solid/open symbols represent downward/upward dipping directions with 95 per cent confidence limits. The star represents the mean direction.

Silurian pole and the Permian one for the NCB is interpreted as indicating the absence of distinct vertical rotation and latitudinal displacement for the NCB during the period Late Silurian to Permian (Meng & Coe 1992). Nevertheless, this provisional palaeomagnetic pole is significantly different from those of Zhao *et al.* (1993) and ours, but is located close to that of the Permian for the NCB (Meng *et al.* 1990). This indicates that the sampling areas in Gansu may have moved independently from Ningxia since the magnetization was acquired (Zhao *et al.* 1996), or the Silurian palaeomagnetic pole of Meng & Coe (1992) might be a result of Late Palaeozoic remagnetization.

The contradiction between our Silurian and Devonian palaeomagnetic poles and those of Zhao *et al.* (1993) requires discussion. Local rotation might not be completely excluded for our Silurian and Devonian palaeopoles and those of Zhao *et al.* (1993), because these poles as well as those of Late Palaeozoic age from the Alashan/Hexi Corridor terrane almost share a small circle arc centred on the sampling site at 116°E, 38°N (Fig. 9). As demonstrated by Wu *et al.* (1993), internal displacement and local deformation in the Alashan/Hexi Corridor terrane probably occurred before the end of the Triassic. Indeed, the Silurian Yezhugou section sampled by Zhao *et al.* (1993) is separated by several tens of kilometres from our Silurian Zhaohuajing sampling section (Fig. 1b). However, the Silurian strata in the Yezhugou section are incomplete, and the Early–Middle Silurian series is in contact with the Early Ordovician Miboshan and the younger Late Silurian Hanxia or Late Devonian Zhongning formations, both with fault contacts (NBGMR 1990). The Late Silurian Hanxia Formation is also in contact with the Ordovician and the younger Late Devonian formations by faults. On the other hand, we have obtained an Early–Middle Ordovician palaeomagnetic pole (326.5°E, 31.8°N, $A_{95} = 9.5$) from the Miboshan area southwest of Ningxia (Huang *et al.* 1999b), which is quite close to the Xiaoyuchuan section of this study (Fig. 1b). This

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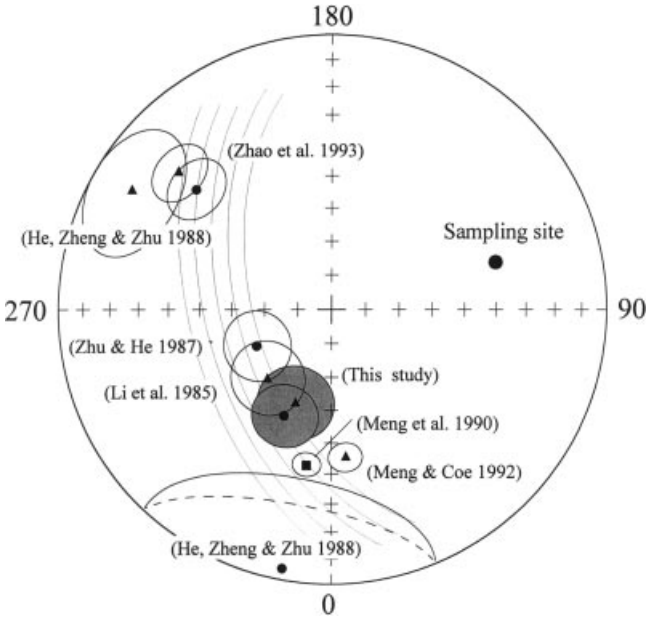


Figure 9. Equal-area projection of the available Palaeozoic palaeomagnetic poles with 95 per cent confidence limits for the Alashan/Hexi Corridor terrane, all plotted onto the Northern Hemisphere. Triangles, circles and squares represent Silurian, Devonian and Permian palaeopoles, respectively.

preliminary palaeomagnetic pole passes fold and reversal tests and shows good consistency with those from the NCB (Zhao *et al.* 1992; Yang *et al.* 1996; Huang *et al.* 1999a), indicating that no relative movement occurred between the Miboshan area of the Alashan/Hexi Corridor terrane and the NCB during that period. In addition, our Silurian and Devonian results from three sampling sections, Zhaohuajing, Shixiagou, and Xiaoyuchuan, show good consistency, indicating that no relative rotation took place among them. Thus, our Ordovician, Silurian and Devonian results obtained from the eastern part of the Hexi Corridor may exclude local rotation and represent the record of an autochthonous terrane.

5.2 Tectonic implications

Together with our preliminary palaeomagnetic pole obtained from Early–Middle Ordovician limestones in Ningxia (Huang *et al.* 1999b) and the available Cambrian and Ordovician palaeomagnetic poles for the NCB (Zhao *et al.* 1992; Yang *et al.* 1996; Huang *et al.* 1999a), two mean poles were calculated for the Cambrian and Ordovician of the NCB (Figs 10a and b, and Table 4). A relatively complete Palaeozoic APWP is then established as shown in Fig. 10(c), by combining our Silurian and Devonian palaeomagnetic results with the mean Cambrian and Ordovician palaeopoles, and the Permian one for the NCB (Enkin *et al.* 1992).

A low palaeolatitude is claimed for the NCB during the Silurian–Devonian period (Table 4), as implied by the Silurian

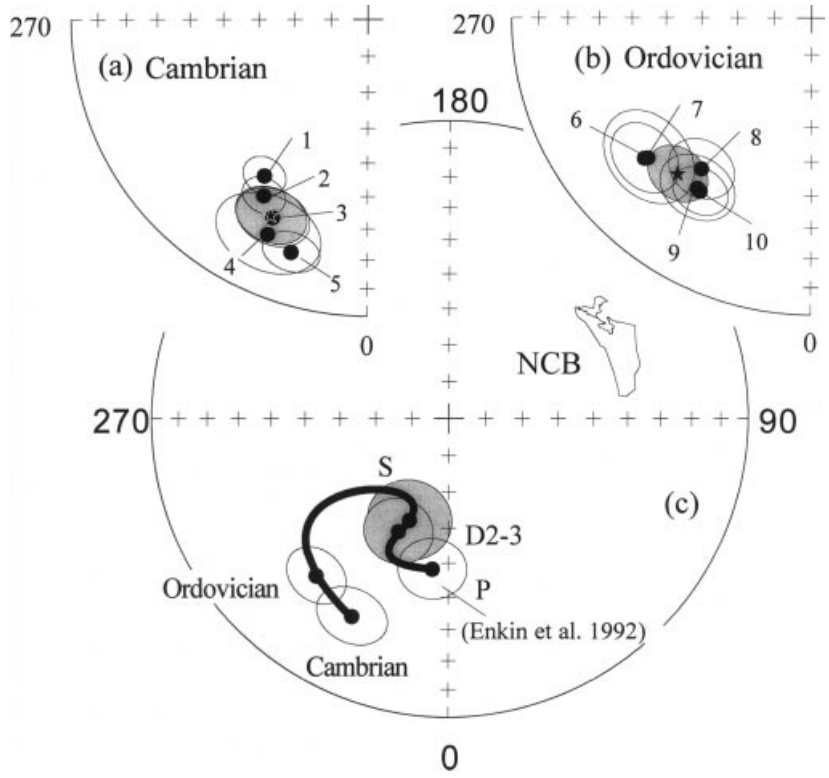


Figure 10. Equal-area projection of the established Palaeozoic APWP for the NCB (c), which is compiled from two mean Cambrian (a) and Ordovician (b) palaeopoles calculated from the available Cambrian and Ordovician palaeopoles for the NCB, the Silurian and Devonian palaeopoles of this study with shaded 95 per cent confidence limits, and the Permian one of Enkin *et al.* (1992). Palaeopoles: 1, 2, 5, 8, and 10 from Huang *et al.* (1999a); 3, Gao *et al.* (1983); 4 and 7, Zhao *et al.* (1992); 6, Yang *et al.* (1996); 9, Huang *et al.* (1999b). All the poles are plotted onto the Northern Hemisphere with 95 per cent confidence limit. S, D and P represent Silurian, Devonian and Permian, respectively.

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Table 4. Available Palaeozoic palaeomagnetic poles for the NCB and its adjacent blocks.

No.	Age	Long (°E)	Lat (°N)	A_{95} (°)	Palaeolat (°)	Reference
North China block(NCB)						
1	Cam2	326.7	37.0	5.5	−6.1	Huang *et al.* (1999a)
2	Cam3	329.6	31.7	5.4	−9.3	Huang *et al.* (1999a)
3	Cam	334.5	26.8	8.9	−10.9	Gao *et al.* (1993)
4	Cam	335.2	21.2	12.4	−14.9	Zhao *et al.* (1992)
5	Cam1	341.9	18.5	6.5	−12.9	Huang *et al.* (1999a)
Mean	Cam	333.9	27.1	8.7	−10.9	
6	O2	310.4	27.9	9.2	−20.3	Yang *et al.* (1996)
7	O1–2	310.9	28.8	12.3	−19.3	Zhao *et al.* (1992)
8	O1	324.3	37.4	8.5	−6.7	Huang *et al.* (1999a)
9	O1–2	326.5	31.8	9.5	−10.7	Huang *et al.* (1999b)
10	O2	327.7	31.5	7.0	−10.4	Huang *et al.* (1999a)
Mean	O1–2	319.8	31.7	7.8	−13.5	
	P	353.9	48.0	8.8	15.0	Enkin *et al.* (1992)
Alashan/Hexi corridor terrane						
	S	339.0	60.1	11.2	17.3	This study
	S	228.4	26.2	8.2	−6.1	Zhao *et al.* (1993)
	S3	317.0	62.0	11.0	13.1	Li *et al.* (1985)
	S	239.7	16.3	(11.3,19.2)	−20.5	He *et al.* (1988)
	S3	5.5	45.1	4.7	19.6	Meng & Coe (1992)
	D2–3	337.0	56.0	9.2	13.1	This study
	D2–3	228.7	34.2	8.8	−0.3	Zhao *et al.* (1993)
	D3	349.6	4.4	(21.1, 37.4)	−17.6	He *et al.* (1988)
	D	296.0	65.3	(6.5, 11.5)	13.6	Zhu & He (1987)
	P2	350.9	42.4	3.9	9.7	Meng *et al.* (1990)
South China block (SCB)						
	S	194.7	4.9	5.6	4.0	Opdyke *et al.* (1987)
	D2	190.4	8.9°S	6.9	−1.1	Fang, *et al.* (1989)
Tarim block (TAM)						
	D	165.0	16.0	4.3	34.0	Li (1990)
	D3	151.2	10.5°S	7.7	25.7	Bai *et al.* (1987)

Abbreviations: Long/Lat, longitude and latitude of palaeopole; Palaeolat, palaeolatitude at reference site of 106.0°E, 38.0°N; A_{95}, radius of circle (semi-axis along and perpendicular to the great-circle of ellipse) of 95 per cent confidence about the palaeopole in degrees. Cam, O, D, S, and P represent Cambrian, Ordovician, Silurian, Devonian, and Permian, with 1 = Early, 2 = Middle, and 3 = Late.

and Devonian palaeomagnetic results for South China and Tarim (Bai *et al.* 1987; Li 1990; Opdyke *et al.* 1987; Fang *et al.* 1989). The Silurian and Devonian palaeomagnetic poles suggest that the NCB was located at around 15°N in the Northern Hemisphere during that period. On the other hand, a 32° clockwise polar wandering between the Early–Middle Ordovician and the Silurian implies that the NCB experienced a large northward latitudinal displacement of $30.8° \pm 10.9°$ and passed the palaeo-equator during that period. The small deviation between the palaeopoles of the Silurian and Devonian may indicate that rapid northward displacement of the NCB in the Palaeozoic had ceased by the Silurian. Insignificant palaeolatitudinal changes would be indicated from the Silurian to Devonian, as well as to the Late Permian for the NCB, which are limited within palaeomagnetic error ($4.2° \pm 11.6°$ for the Silurian to Devonian and $-1.9° \pm 10.2°$ for a period from Devonian to Late Permian). Moreover, our Silurian and Devonian palaeomagnetic poles yield insignificant or little vertical axis rotation for the NCB from the Middle Ordovician to the Late Permian, which seems to be consistent with the deduction that no distinct vertical axis rotation occurred for the NCB during the Late Palaeozoic (Meng & Coe 1992).

5.3 Palaeogeographical reconstruction for the Late Devonian

The NCB was possibly situated close to Australia during the Late Devonian. It should be noted that the occurrence of *Sinolepid antiarches*, a group of endemic freshwater fish fossils found in the Upper Devonian strata of Ningxia and reported only from Australia and South China (Pan *et al.* 1987; Burrett *et al.* 1990), is evidence not to be ignored for the Devonian palaeogeographical discussion of the NCB. The NCB is consequently inferred to be in the proximity of Australia in the Late Devonian (Fig. 11). This reconstruction is similar to that of Long & Burrett (1989), deduced mainly from the palaeontological and palaeogeographical evidence. On the other hand, this configuration also correlates the Silurian–Devonian segment of the APWP for the NCB with that for Gondwana (Fig. 11), indicating that the NCB was part of East Gondwana during that period. East Gondwana was glaciated at southern polar latitudes in the Late Palaeozoic, whereas the NCB remained at low latitudes during that period (Zhao *et al.* 1996). The timing of the rifting of the NCB off East Gondwana may therefore be constrained to have taken place after the Devonian.

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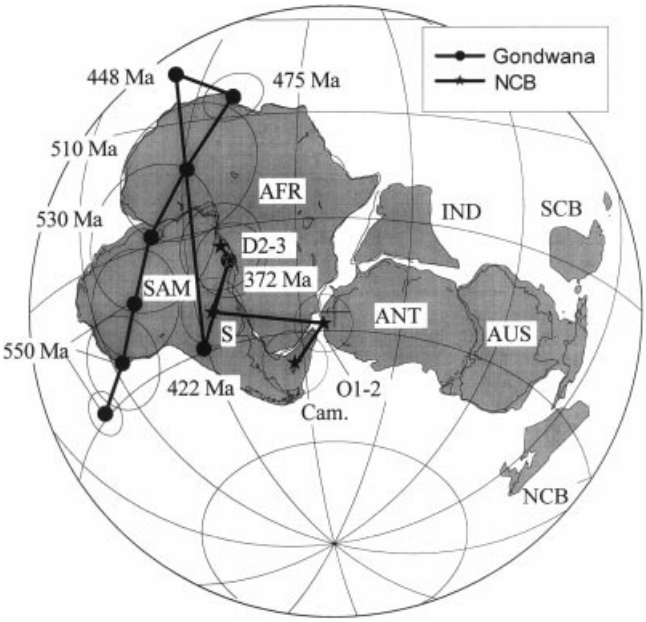


Figure 11. Schematic map showing the reconstruction of the NCB with neighbouring regions of the world in the Late Devonian in African coordinates. Relative longitudes of the blocks are unconstrained. The APWP for Gondwanaland (circles) is drawn according to Meert & Van der Voo (1997), Grunow (1995), and Van der Voo (1993). All the poles are plotted with palaeo-south poles. Abbreviations are: AFR, Africa; ANT, Antarctica; AUS, Australia; IND, India; NCB, North China Block; SAM, South America; SCB, South China Block. The NCB is shown relative to Africa according to a rotation pole of 0.8°N, 270.3°E with $\Delta = 119.6$ whereas Gondwanaland is shown as in Huang *et al.* (1999a). See text for further discussion.

6 CONCLUSIONS

A high-temperature characteristic component B was isolated from most samples of Silurian and Devonian red beds, marlaceous sandstone, and a few limestone rocks from widely distributed sections in the eastern part of the Hexi Corridor, southwest Ningxia in North China. We conclude that this component is primary on the basis of positive fold and reversal tests.

We presume here that the Hexi Corridor terrane has formed an integral unit block with the NCB since the Late Cambrian. The Palaeozoic APWP for the NCB is established from the Silurian and Devonian palaeomagnetic poles of the Hexi Corridor and two mean Cambrian and Ordovician ones from the NCB. The newly established Palaeozoic APWP indicates that the NCB was situated at around 15°N in the Northern Hemisphere during the Silurian–Devonian period and had some affinities with East Gondwana during that period.

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